14.6 A 28nm 64kb Bit-Rotated Hybrid-CIM Macro with an Embedded Sign-Bit-Processing Array and a Multi-Bit-Fusion Dual-Granularity Cooperative Quantizer

Xi Chen, Shaochen Li, Zhican Zhang, Wentao Zheng, Xiao Tan, Yuchen Tang, Yuhui Shi, Lizheng Ren, Yibo Mai, Feiran Liu, Jinwu Chen, Zhaoyang Zhang, An Guo, Tianzhu Xiong, Bo Wang, Xinning Liu, Weiwei Shan, Bo Liu, Hao Cai, Jun Yang, Xin Si

Southeast University, Nanjing, China

Hybrid-domain CIMs [1-3] are attracting increasing attention nowadays as they combine the advantages of both digital CIMs (DCIM) [4-6] and analog CIMs (ACIM) [7-10], offering a more balanced choice. As depicted in Figure 14.6.1, when designing a hybrid CIM (HCIM), the first challenge to address is (1) the definition of boundary between the digital and analog part, which is determined by feature input mode and weight mapping. Prior HCIMs either utilized a bit-parallel in features to obtain high accuracy at the cost of hardware (HW) overhead [1] or realized a low HW overhead but suffered from accuracy loss due to error pollution problem (error of the analog partial sum makes the high precision digital partial sum of the same bit weight meaningless) with bit-serial scheme and specific weight mapping strategy [2,5], neither of which provided a perfect solution to the boundary question. Therefore, we propose a bit-rotated feature-in scheme to address this challenge. To compress multiple multiplication results into one MAC result, several levels of shifters and adders are usually required, and the sign-bit needs to be considered when compressing signed and unsigned information. Unlike the bit-parallel/serial approach [1,2], the bit-rotated scheme considers the sign-bit in the first product-wise adder, which induces (2) more digital overhead for sign-bit processing. Another challenge that continues to plague bit-rotated HCIM is (3) excessive energy wasted on low accuracy-contributed low-bit quantization in analog parts.

The contributions of this work are as follows: (1) a hybrid structure with a bit-rotated feature-in scheme for low HW overhead and high accuracy; (2) an embedded sign-bit-processing (ESP) hybrid SRAM array for correct and low HW overhead computation; (3) a multi-bit-fusion dual-granularity cooperative quantizer (MF-DGCQ) for multi-bit multi-cycle low-power/delay quantization. The proposed 28nm 64kb bit-rotated hybrid-CIM macro achieves a high, 67.8TOPS/W, energy efficiency for INT8 MAC operations and represents the most balanced point of HCIM.

Figure 14.6.2 presents the overall structure and operations of the proposed bit-rotated hybrid-CIM macro. The overall structure consists of a 256 × 256b ESP hybrid SRAM array, a gather and rotate input buffer (GRIB), WL drivers, a controller (CTRL), write IO (WIO), an MFC-DGCQ, and a post processing unit (PPU). The ESP hybrid SRAM array consists of 16 channel-wise MAC units (CWMAUs). Each CWMAU includes 32 product-wise MAC units (PWMAUs) and a channel-wise adder (CWA). A sign-bit MAC cell (SBMC), 7 hybrid-MAC cells (HBMCs) and a product-wise adder (PWA) form a PWMAU. During GEMM, static weights are preloaded to the SRAM array, with the most significant bit (MSB) stored in the SBMC while other bits are in HBMCs. Features are fetched by the gather buffer (GB) and written to the array via GBL/GBLBs by the rotated-input buffers (RIUs) using the proposed bit-rotated method to perform product-wise first hybrid MAC operations, considering sign-bit processing in 8 cycles (take INT8 MAC as example). Digital MAC results, as well as the sign-bit extension, are accumulated by the PWA and CWA, while the analog part is computed in the charge domain; the output results are quantized by MFC-DGCQ. The two results are combined by the PPU. Thus, a vertical-cut hybrid pattern is achieved without using a bulky MAC array and digital adders.

Figure 14.6.3 illustrates detailed schematics and operations of the GRIB and ESP hybrid SRAM array. Each GB consists of 8 DFFs and is responsible for gathering serial input data and delivering them to the 8 corresponding RIUs in parallel. Each RIU comprises of two 2:1 MUXs and a DFF that work in fetch mode or rotate mode. A complete rotated-MAC operation occupies one fetch period and 7 rotate periods. During the fetch period (rot = 0), 8b features and 8 zeros are fetched simultaneously from the GB by the left and right DFFs of 8 RIUs. During the rotate period (rot = 1), features are transferred to adjacent DFFs within the left and right groups; particularly, feature data from the left DFF of RIU#7 will be transferred to the right DFF of RIU#0, and 0 will be transferred to the left DFF of RIU#0. There are two kinds of MAC cells in the ESP hybrid SRAM array: (1) the SBMC comprises of 16 6T-SRAM cells, a selective reverse digital multiplier (SRDM) and a sign-bit extension unit (SBEU); (2) the HBMC comprises of 16 6T-SRAM cells, an SRDM and an analog multiplier (AM). During each cycle, a 1b weight is read from one of the 16 6T-SRAM cells and multiplied by 2b features on GBL/GBLB; the result from the SRDM contributes to the digital part, while the one from the AM contributes to the analog part. The ESP hybrid SRAM array supports both unsigned and signed modes. When it works under signed mode, two aspects of sign-bit processing need to be considered: (1) selective reversion, where the SRDM would deliver the original multiplication results or the reversed ones, depending on whether F[7] is the multiplier or not; (2) sign-bit extension, where previous results (p07, p17, etc.) would be stored and reused by the 3b register and adder (3b_RA) in the SBEU. While in unsigned mode, the SRDM would deliver the original multiplication results and SBEU does not perform sign-bit extension.

Figure 14.6.4 shows the proposed MF-DGCQ for multi-bit multi-cycle low-power/delay quantization. In our proposed HCIM, an 8b ADC is utilized to quantize the analog MAC result; the accumulation length is set to 32 per cycle for each CWMAU. Conventional quantization schemes activate the ADC every cycle, resulting in a poor output ratio (OR: ratio of practical output resolution to the ideal output range) for low-bit quantization. The proposed multi-bit-fusion quantization scheme quantizes multiple low bits together to reduce the ADC activation time. The multi-bit-fusion circuit (MFC) comprises of 3 sets of write (WR) and read (RD) switches, 3 fusion capacitors and a compensation capacitor, whose capacitance ratio is 1:2:4:1. In the reset phase (period#0), all data on GBLB is 0, which means the upper plate of the computation capacitor in the PWMAU is connected to 0. Meanwhile, all switches in the MFC are closed, to connect the bottom plates of the computation capacitors, the upper plates of fusion capacitors, and the compensation capacitor to 0. In the write phase, coupling voltages V_c that represent MAC results are obtained on the fusion capacitors successively. In the read phase, corresponding RD switches are closed to perform charge sharing between the fusion and compensation capacitors; the obtained sharing voltage, V_s, represents the multi-bit MAC result. V_s is quantized by the DGCQ, and then reset to 0. The proposed DGCQ consists of a coarse-grained and a fine-grained part. The coarse-grained quantizer (CGQ) comprises a comparator, a 4b counter, and a voltage subtraction circuit (VSC). The fine-grained quantizer (FGQ) is an 8b SAR-ADC. When V_s from the MFC is ready, it is transported to the DGCQ and performed charge sharing with the prior accumulation voltage V_{acc}. In most quantization cycles (when the CGQ activated), the new V_{acc} is compared with V_{cm} ($^1/_4V_{DD}$) to determine the behavior of the counter and VSC. The counter records the times when $V_{acc} > V_{cm}$, and the VSC regulates V_{acc}. Once multi-cycle computation is done, the FGQ is activated to quantize the remaining V_{acc}. Both coarse-grained and fine-grained results are delivered to the PPU to obtain the final MAC result.

Figure 14.6.5 presents the simulation results of the proposed HCIM macro. We evaluate our proposed bit-rotated structure on compute error by performing a 10k Monte Carlo simulations. Compared to prior hybrid structures [1,2,3], this work outperforms the most advanced bit-serial structures. From the perspective of HW, this work outperforms typical bit-parallel/serial work [1,2] on the FoM (area efficiency × energy efficiency) by 15.7× and 3.38×. The MF-DGCQ allows this work to achieve a 24% performance enhancement and a 4.27× power reduction with a relative error of 3.57% in the typical process corner, which occurs a negligible accuracy loss.

As shown in Fig. 14.6.6, the bit-rotated HCIM macro is fabricated using 28nm CMOS technology. This work achieves a decent accuracy on emerging AI applications: accuracy loss is -1.06% for ResNet-18@ImageNet; -1.75% for ViT@ImageNet; 0.19 for GPT-2@Wikitext-102. For INT8 MAC operations with 32 channel accumulations, the measured access time is 12ns at 0.9V. The peak-energy and area efficiency are 67.8TOPS/W and 1.57TOPS/mm^2. By leveraging a bit-rotated feature-in scheme, this work proposes the most balanced hybrid computing structure, improving the memory density × area efficiency × energy efficiency FoM by more than 28% in comparison to prior SRAM-CIM macros [1-3,5,7,9]. Figure 14.6.7 shows the die micrograph and a chip summary table.

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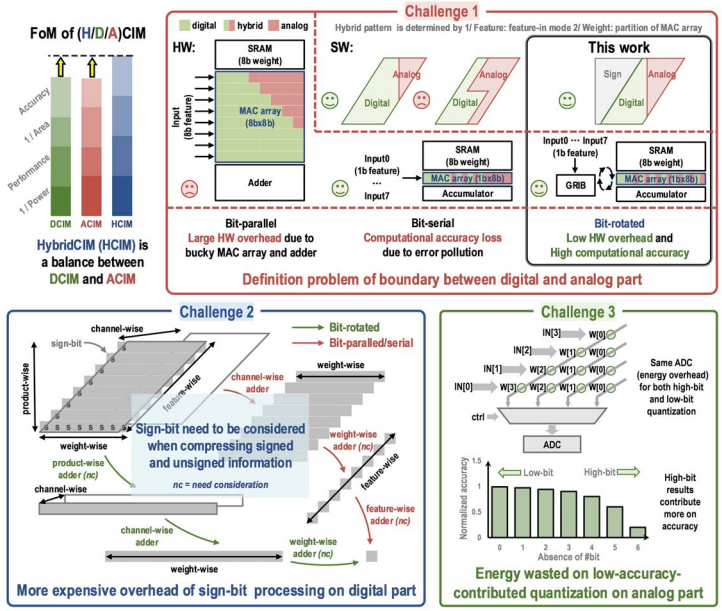


Figure 14.6.1: Hybrid CIM challenges.

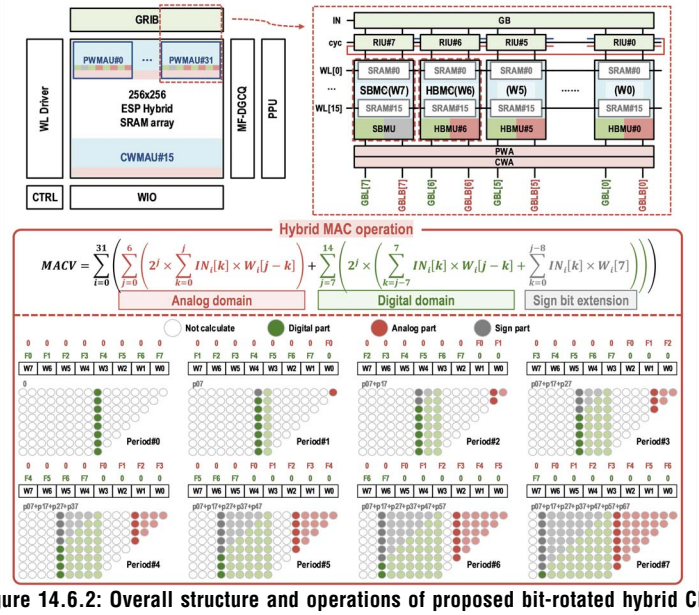


Figure 14.6.2: Overall structure and operations of proposed bit-rotated hybrid CIM macro.

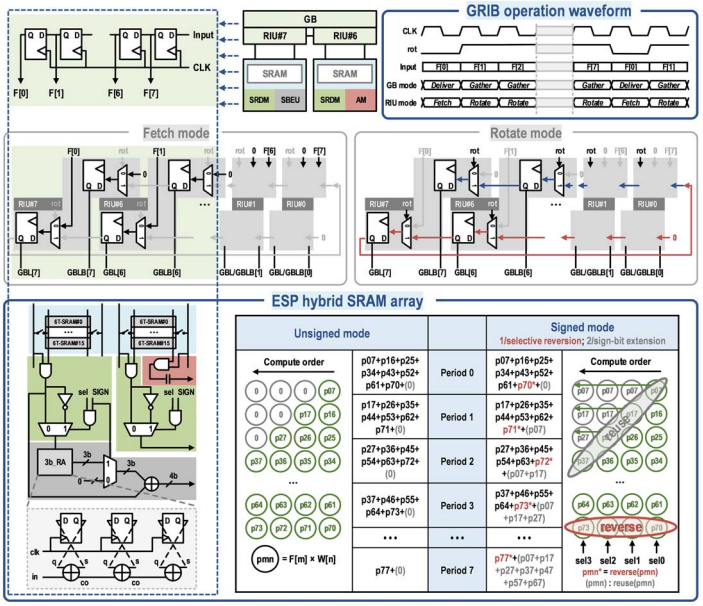


Figure 14.6.3: Detailed schematic and operations of GRIB and ESP hybrid SRAM array.

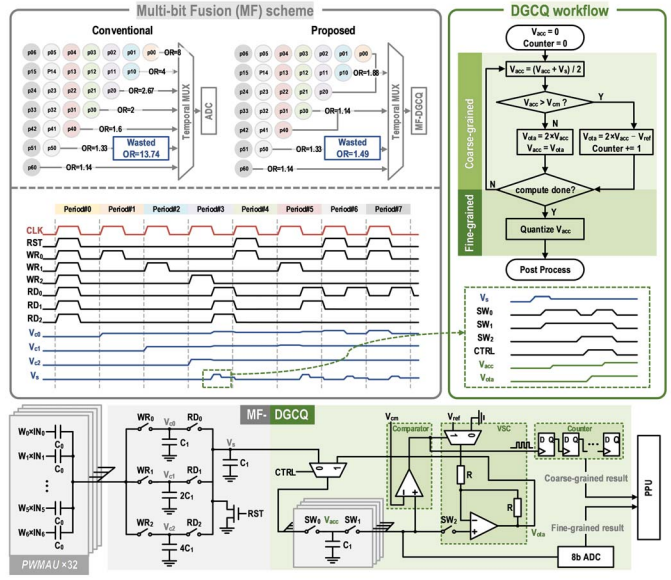


Figure 14.6.4: MF-DGCQ schematic and operations for multi-bit multi-cycle low-power/delay low-bit quantization.

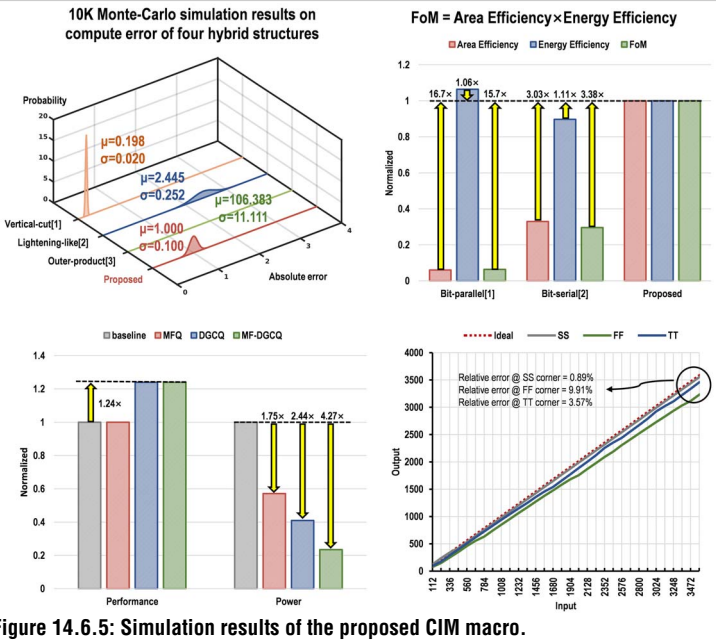


Figure 14.6.5: Simulation results of the proposed CIM macro.

Model	ResNet-18	ViT	GPT-2
Dataset	ImageNet	ImageNet	Wikitext-102
Data precision	INT8	INT8	INT8
Task	Classification	Classification	NLP
Metric	Accuracy	Accuracy	Perplexity
Baseline	69.54%	80.24%	48.39
Proposed	68.48%	78.49%	48.58

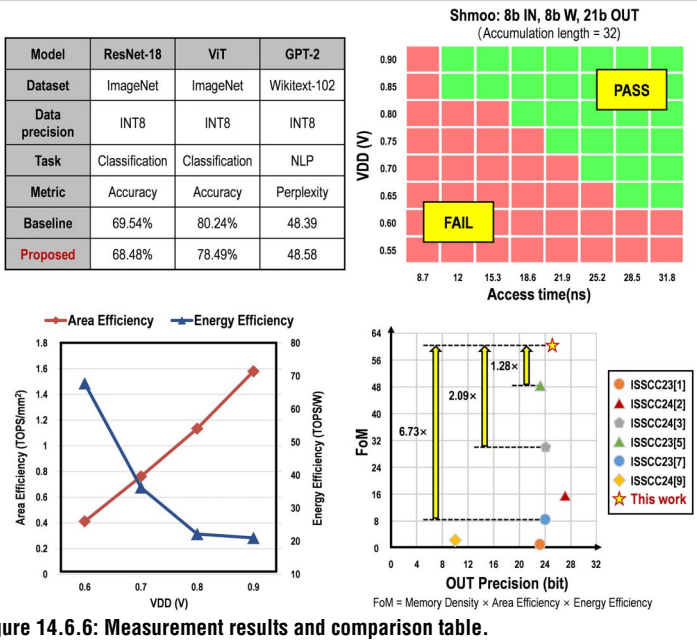
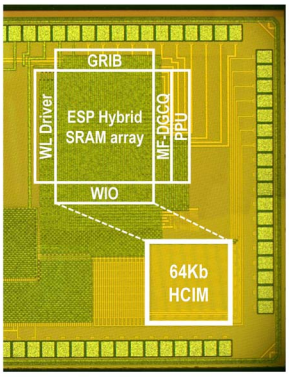


Figure 14.6.6: Measurement results and comparison table.

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[1]Measured with worst case under 0.9V. Worst case: 50% feature sparsity, 0% weight sparsity.
[2]Measured under 0.6V.
[3]Software baseline is 69.54%.
[4]Software baseline is 80.24%.
[5]Software baseline is 48.39.

CHIP SUMMARY		
Technology		28nm
Bitcell		6T compact
MAC Domain		Hybrid
Macro Capacity		256×256 (64Kb)
Macro Area (mm^2)		330.19um×328.35um =0.108mm^2
Memory Density (Mb/mm^2)		0.58
Supply Voltage (V)		0.6-0.9
Input precision (bit)		8
Weight precision (bit)		8
Output precision (bit)		21-25
Accumulation length		32-512
Access Time (ns)		12@0.9V
Throughput (TOPS)		[2]0.04-[1]0.17
Area Efficiency ($TOPS/mm^2$)		[2]0.37-[1]1.57
Energy Efficiency (TOPS/W)		[1]21.04-[2]67.8
Inference Accuracy / Perplexity	[3]ResNet18@ImageNet	-1.06%
	[4]ViT@ImageNet	-1.75%
	[5]GPT-2@Wikitext-102	0.19

Figure 14.6.7: Die micrograph and chip summary table.

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